AI-POWERED EMPLOYEE ATTRITION & WORKFORCE PLANNING

Excel Dataset Explanation and Project Plan

Purpose: This document explains the dataset in simple language, what each column can be used for, which new calculated columns can be created, and what additional data is needed for attrition prediction.

1. Project Objective

The project aims to understand workforce distribution across departments, study employee exits, support vacancy and hiring plans, and—when detailed employee-level data is available—explore machine-learning methods to estimate attrition risk.

- Data Analysis: summarise and compare workforce data.
- Attrition Analysis: study employee exit patterns.
- Workforce Planning: compare current headcount with planned headcount and vacancies.
- Reporting: communicate findings through Excel, SQL, and Power BI dashboards.
- Machine Learning: build an attrition prediction model only if suitable employee-level records and an Attrition target column are available.

2. Workbook Structure

The uploaded workbook was checked. The sheets and column headers found are:

Workforce_Planning

Columns: Department, Total_Employee_Records, Active_Headcount, Notice_Period_Count, Relieved_Records, Target_Headcount_Illustrative, Open_Vacancies_Illustrative, Planning_Note

README

Columns: Read_Me

Important: The exact sheet names and headers above are taken from the uploaded workbook. The explanations below focus on the workforce-planning fields discussed for this project; if the workbook has additional sheets or columns, they should be reviewed before finalising the analysis.

3. Column-by-Column Explanation

Column	Simple meaning	How it helps the project
Department	Department or business unit	Compare headcount, exits, and vacancies by department.
Total_Employee_Records	Total employee records counted for the department	Use as a record-count baseline; confirm the counting rules.
Active_Headcount	Employees currently active, according to the dataset	Measure current workforce and compare departments.
Notice_Period_Count	Employees recorded as being in their notice period	Monitor potential upcoming exits; this is not the same as confirmed attrition.
Relieved_Records	Records of employees who have left / been relieved	Study exits by department; check the time period and whether records are unique

		employees.
Target_Headcount_Illustrative	Illustrative planned workforce target	Compare the current headcount with the planned target; confirm that the target is approved.
Open_Vacancies_Illustrative	Illustrative number of open positions	Support recruitment prioritisation; confirm whether this is an actual vacancy count.
Planning_Note	A note about the department's planning assumptions	Explain caveats and assumptions in reports.

4. What Analysis Can We Do?

Analysis	What it tells us	Suggested output
Department-wise workforce	Compare active headcount across departments.	Bar chart; headcount summary table.
Employee exit pattern	Compare relieved records by department.	Bar chart; verify the period and employee denominator.
Notice-period monitoring	See where more employees are currently in notice period.	Department-wise notice-period chart.
Workforce gap	Compare active headcount with the target headcount.	Gap table and conditional formatting.
Vacancy planning	Compare open vacancies across departments.	Vacancy ranking and recruitment priority list.
Management dashboard	Bring the main measures together.	Power BI KPI cards, bar charts, filters, and tables.

5. Useful Calculated Columns

These can be created in Excel or SQL after confirming the source definitions.

New column	Calculation / rule	Purpose
Relieved_Record_Share_Pct	Relieved_Records / Total_Employee_Records × 100	Share of records marked as relieved. This is not necessarily the official attrition rate.
Notice_Period_Share_Pct	Notice_Period_Count / Total_Employee_Records × 100	Share of records in notice period.
Headcount_Gap	Target_Headcount_Illustrative − Active_Headcount	Positive = below target; negative = above target.
Workforce_Status	If gap > 0: Below Target; if gap = 0: At Target; if gap < 0: Above Target	A simple label to make the dashboard easier to read.
Vacancy_Status	If open vacancies > 0: Vacancies Open; otherwise: No Open Vacancies	Helps flag departments that may need recruitment follow-up.

6. Example Excel Formulas

Purpose	Excel formula (Excel Table)	Note
Relieved record share	=IFERROR([@Relieved_Records]/[@Total_Employee_Records]*100,0)	Format as percentage or numeric percentage consistently.
Notice-period share	=IFERROR([@Notice_Period_Count]/[@Total_Employee_Records]*100,0)	This is a share of records, not an attrition rate.
Headcount gap	=[@Target_Headcount_Illustrative]-[@Active_Headcount]	Positive gap means the active count is below the target.
Workforce status	=IF([@Headcount_Gap]>0,"Below Target",IF([@Headcount_Gap]=0,"At Target","Above Target"))	Requires the Headcount_Gap column to exist.

7. Additional Columns Needed for Employee-Level Attrition Analysis

The current department summary is useful for workforce planning, but it does not show individual employee characteristics. To analyse likely drivers of attrition or train a machine-learning model, request a properly anonymised employee-level dataset.

Suggested column	Why it may be useful
Employee_ID	A unique, anonymised employee key; avoid names and personal contact details.
Department	Employee's department.
Job_Role / Job_Level	Role and seniority level.
Age_Group or Age	Use only where lawful, necessary, and approved.
Job_Satisfaction	Survey score, with its scale documented.
Work_Life_Balance	Survey score or defined measure.
Monthly_Income or Salary_Band	Compensation measure, access-controlled.
OverTime	Whether overtime is recorded, with a clear definition.
Years_At_Company	Tenure in the organisation.
Years_Since_Last_Promotion	Time since the last promotion.
Training_Hours	Training received during a defined period.
Performance_Rating	Performance measure and rating scale.
Attrition	Target label such as Yes/No, based on a documented exit definition.
Snapshot_Date / Exit_Date	Dates needed to define observation periods and avoid data leakage.

8. Important Data Quality and Privacy Checks

- Check whether employee records are unique and whether the same employee can appear more than once.
- Confirm the reporting period for active headcount and relieved records.
- Verify whether illustrative target headcount and vacancy values are planning examples or approved business figures.
- Check missing values, duplicate rows, invalid values, and inconsistent department names.
- Use anonymised employee IDs; do not include names, phone numbers, addresses, or other unnecessary personal identifiers.
- For machine learning, use only information available at the prediction date. Do not use future exit information as a predictor.
- Review model performance and fairness across groups; treat predictions as decision support, not as the sole basis for employment decisions.

9. Suggested Tools and Workflow

Step	Tool	Activity
Step 1	Excel	Inspect columns, clean data, create calculated columns, and validate totals.
Step 2	SQL	Group and compare departments, headcount, relieved records, and vacancies.
Step 3	Power BI	Create KPI cards, department charts, filters, and a workforce planning dashboard.
Step 4	Python	Explore employee-level data and prepare a machine-learning dataset, if available.
Step 5	Model evaluation	Evaluate accuracy, recall, precision, and fairness; document limitations.

10. Professional Explanation for the Team Leader

Our project focuses on analysing employee workforce data to understand department-wise workforce distribution, employee exits, notice-period counts, and open vacancies. The current workbook can support summary analysis and workforce planning by comparing active headcount with illustrative targets. We can create calculated fields such as relieved-record share, notice-period share, headcount gap, and workforce status, then present the results in a Power BI dashboard. To study attrition drivers and train a machine-learning model, we will need anonymised, employee-level data with a clearly defined Attrition label and reporting dates. We will validate the data and business definitions before presenting conclusions.

11. Recommended Next Steps

1. Confirm the meaning and reliability of every source column with HR or the dataset owner.
2. Validate headcount, relieved-record, target, and vacancy totals.
3. Create the calculated columns and build a department-level dashboard.
4. Request the detailed employee-level dataset if attrition prediction is part of the project scope.
5. Document definitions, assumptions, data privacy measures, and model limitations.